

# Distributed Deep Learning on GPU-based Clusters

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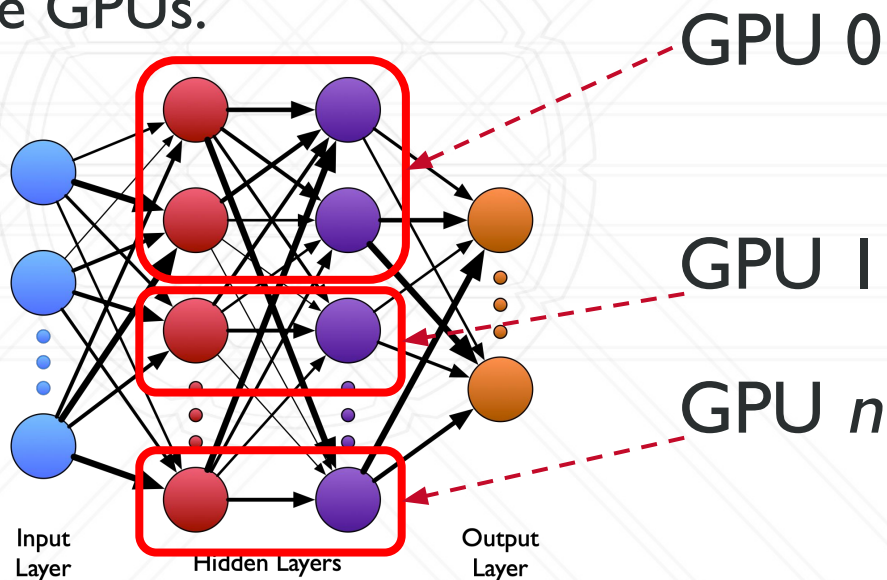
# Limitations of data parallelism

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- DDP – Supports models of limited size
- FSDP – Large communication overheads, specially for small batch size tasks like IFT

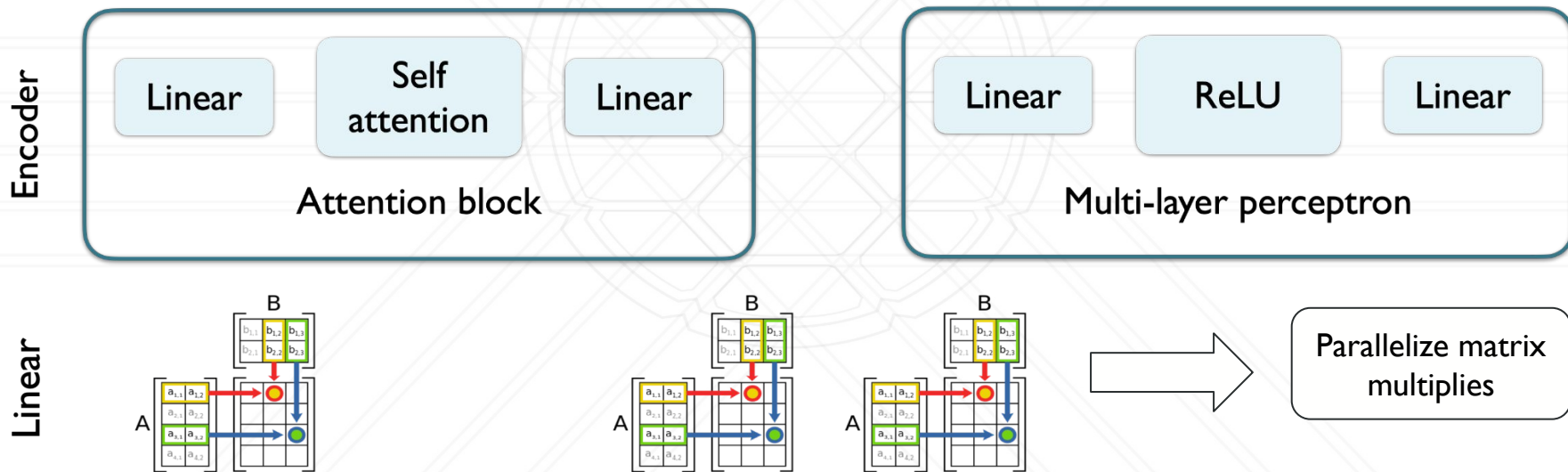
# Tensor parallelism

- Divide parameters and compute of every layer of a neural network on multiple GPUs.



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# Intra-layer parallelism

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- Divide parameters and compute of every layer of a neural network on multiple GPUs.
- Two kinds of layers
  - ReLU and Layernorm – apply same function to each element of the input tensor
  - Fully Connected/Convolution – matrix multiplications

# Creating an AxoNN Lightning Strategy

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```
from axonn.lightning import AxonnStrategy
```

```
pl_strategy = AxonnStrategy(
```

```
    G_intra_r=..
```

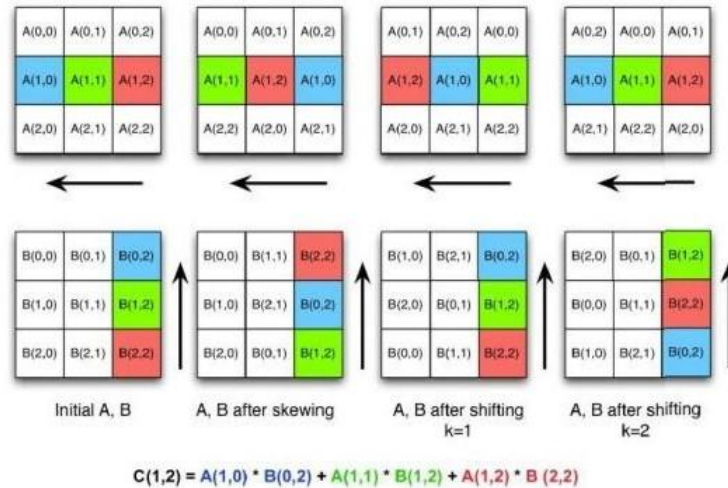
```
    G_intra_c=..
```

```
    G_intra_d=..
```

```
)
```

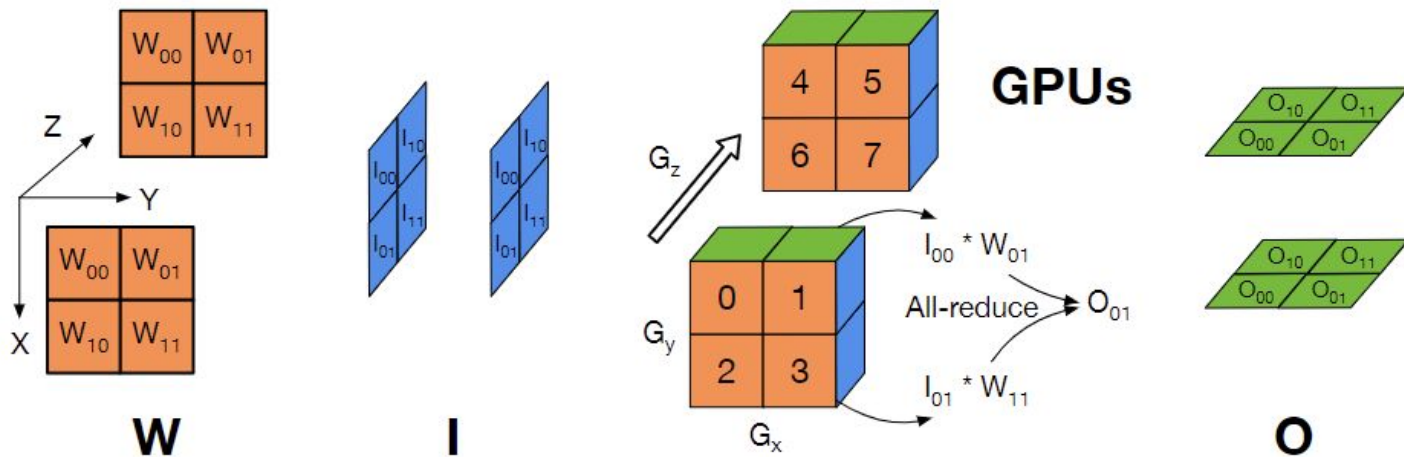
3D tensor parallel grid  
dimensions

# Parallelizing a Matrix Multiplication



Cannon's Parallel Matrix Multiplication Algorithm

# AxoNN's 3D Tensor Parallelism



Parallelizing a matrix multiplication ( $I.W=O$ ) using AxoNN on 8 GPUs



# Running the code (Tensor/Intra-Layer)

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- Code – `train.py`

```
CONFIG_FILE=configs/axonn.json sbatch train_multi.sh
```

# Let's try different AxoNN configurations

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- In config/axonn.json - tp\_dimensions is [2, 2, 1].
- Now change it to [4, 1, 1] and rerun your code.
- Does it become faster?

# How to set the parallel configuration?

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- Small batch-sizes - Use  $x$  and/or  $y$ 
  - Example - finetuning and inference
- Large batch sizes - Use  $z$  (+data parallelism)
  - Example - pretraining

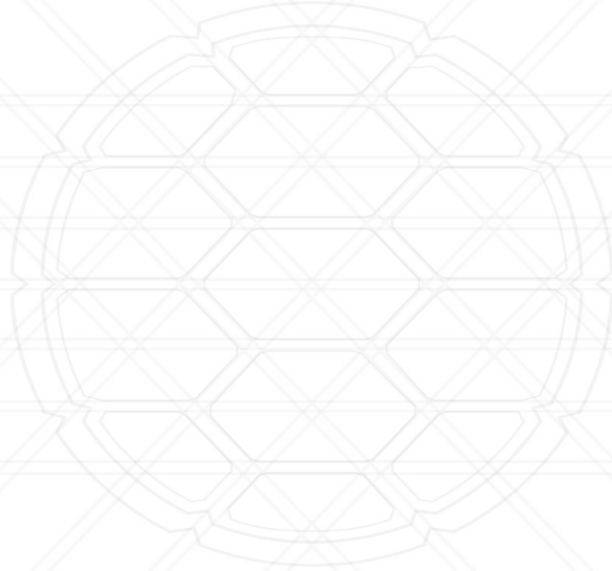
# Alternative way to use AxoNN

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```
from axonn.intra_layer import auto_parallelize
with auto_parallelize():
    net = FC_Net(args.num_layers, args.image_size**2, args.hidden_size, 10).cuda()
```

Zero code changes required in your model definition!

`fabric.init_module()` calls AxoNN's auto-parallelize function



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